

Contest Mindset and Diagnostic

Entrance Diagnostic · Form A

Name		Date	
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60 minutes. No calculator. 25 questions in three parts. Questions 1–10 are worth 2 points, 11–20 are worth 5 points and 21–25 are worth 6 points (100 points in total). For questions 1–20, circle one answer and write its letter on the line or in the box. For questions 21–25, write your answer on the answer line. Where you see a method box, show your diagram, table or equation: it can earn points even if your final answer is wrong. Diagrams in this paper are drawn to scale. Skip a question if you are stuck and come back to it.

Part 1 — Foundations. Questions 1–10, 2 points each.

1. What is $\frac{3}{4}$ of 36?
 (A) 9 (B) 12 (C) 24 (D) 27 (E) 48

2. What is 15% of 80?
 (A) 8 (B) 12 (C) 15 (D) 16 (E) 65

3. If $4n + 7 = 43$, what is the value of n ?
 (A) 4 (B) 7 (C) 9 (D) 12.5 (E) 36

4. The pattern 5, 12, 19, 26, ... continues by adding 7 each time. What is the 10th number in the pattern?
 (A) 50 (B) 61 (C) 65 (D) 68 (E) 75

5. A rectangle is 12 cm long and 7 cm wide. What is its perimeter?
(A) 19 cm (B) 26 cm (C) 38 cm (D) 84 cm (E) 168 cm
-
6. Two angles of a triangle measure 48° and 67° . What is the measure of the third angle?
(A) 45° (B) 55° (C) 65° (D) 115° (E) 295°
-
7. What is the mean (average) of 8, 11, 12, 15 and 24?
(A) 11 (B) 12 (C) 13 (D) 14 (E) 70
-
8. Which of the following is a prime number?
(A) 59 (B) 39 (C) 51 (D) 57 (E) 91
-
9. What is the lowest common multiple of 6 and 8?
(A) 2 (B) 12 (C) 14 (D) 16 (E) 24
-
10. Ana, Ben and Cy each own exactly one pet: a cat, a dog or a fish, all different. Ben does not own the dog. Cy owns the cat. Who owns the fish?
(A) Ben (B) Ana (C) Cy (D) Ana or Ben (E) It cannot be determined
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Part 2 — Problem solving. Questions 11-20, 5 points each.

11. A jacket costs \$80. It is on sale for 25% off. Then 13% tax is added to the sale price. What is the final cost?

(A) \$60.00 (B) \$67.80 (C) \$70.40 (D) \$80.00 (E) \$90.40

Method box — a correct diagram, table or equation earns points even if the answer is wrong.

Answer: _____

12. The ratio of red marbles to blue marbles in a bag is 3 : 5. There are 48 marbles. How many more blue marbles than red marbles are there?

(A) 6 (B) 12 (C) 18 (D) 30 (E) 48

Method box — a correct diagram, table or equation earns points even if the answer is wrong.

Answer: _____

13. Mia ate $\frac{1}{3}$ of a pizza. Leo then ate $\frac{1}{4}$ of what was left. What fraction of the whole pizza remains?

(A) $\frac{1}{6}$ (B) $\frac{5}{12}$ (C) $\frac{1}{2}$ (D) $\frac{7}{12}$ (E) $\frac{2}{3}$

Answer: _____

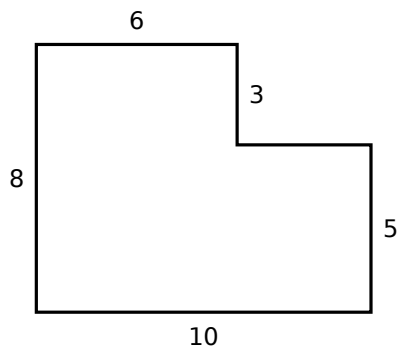
14. Squares are built in a row from toothpicks: 1 square uses 4 toothpicks, 2 squares use 7, and 3 squares use 10. How many toothpicks are needed for 20 squares in a row?

(A) 60 (B) 61 (C) 63 (D) 80 (E) 81

Method box — a correct diagram, table or equation earns points even if the answer is wrong.

Answer: _____

15. The figure is made of rectangles and all corners are right angles. Some side lengths, in cm, are shown. What is the area of the figure, in cm^2 ?

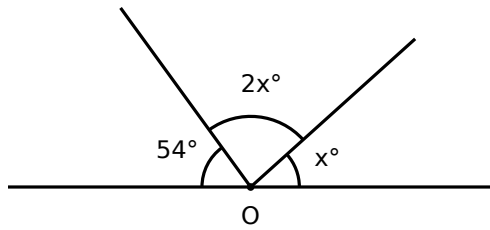


(A) 56 (B) 62 (C) 68 (D) 74 (E) 80

Method box — a correct diagram, table or equation earns points even if the answer is wrong.

Answer: _____

16. In the diagram, O lies on a straight line. Find the value of x .



- (A) 21 (B) 42 (C) 63 (D) 84 (E) 126

Answer: _____

17. A rectangular prism has a volume of 360 cm^3 . Its base is 8 cm by 9 cm. What is its height?

- (A) 3 cm (B) 4 cm (C) 5 cm (D) 20 cm (E) 45 cm

Answer: _____

18. A bag has 4 red marbles, 5 blue marbles and some green marbles. The probability of picking a blue marble is $\frac{1}{3}$. How many green marbles are in the bag?

- (A) 2 (B) 3 (C) 5 (D) 6 (E) 15

Method box — a correct diagram, table or equation earns points even if the answer is wrong.

Answer: _____

19. How many whole numbers from 1 to 100 are multiples of 3 or multiples of 5 (or both)?

(A) 33 (B) 41 (C) 47 (D) 53 (E) 58

Method box — a correct diagram, table or equation earns points even if the answer is wrong.

Answer: _____

20. Four friends finish a race with no ties. Ari finishes before Bo. Bo finishes before Dee. Cal finishes immediately after Dee. Who finishes third?

(A) Ari (B) Bo (C) Cal (D) Dee (E) It cannot be determined

Answer: _____

★ Challenge

Part 3 — Challenge. Questions 21–25, 6 points each. Write your answer on the line.

21. In a club, the ratio of boys to girls is 3 : 4. After 6 more boys join (and nobody leaves), the numbers of boys and girls are equal. How many members does the club have now?

Method box — a correct diagram, table or equation earns points even if the answer is wrong.

Answer: _____

22. The sum of five consecutive even numbers is 170. What is the largest of the five numbers?

Method box — a correct diagram, table or equation earns points even if the answer is wrong.

Answer: _____

23. A $4 \times 4 \times 4$ cube is built from 64 small cubes. The outside of the large cube is painted. How many small cubes have paint on exactly two faces?

Method box — a correct diagram, table or equation earns points even if the answer is wrong.

Answer: _____

24. Two fair six-sided dice are rolled. What is the probability that the two numbers rolled add to 8?

Method box — a correct diagram, table or equation earns points even if the answer is wrong.

Answer: _____

25. How many three-digit whole numbers have digits that add to 5?

Method box — a correct diagram, table or equation earns points even if the answer is wrong.

Answer: _____

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Answers and points

Q	Strand	Points	Answer	Q	Strand	Points	Answer
1	N	2	(D) 27	14	A	5	(B) 61
2	N	2	(B) 12	15	G	5	(C) 68
3	A	2	(C) 9	16	G	5	(B) 42
4	A	2	(D) 68	17	G	5	(C) 5 cm
5	G	2	(C) 38 cm	18	D	5	(D) 6
6	G	2	(C) 65°	19	T	5	(C) 47
7	D	2	(D) 14	20	L	5	(D) Dee
8	T	2	(A) 59	21	N	6	48
9	T	2	(E) 24	22	A	6	38
10	L	2	(A) Ben	23	G	6	24
11	N	5	(B) \$67.80	24	D	6	5/36
12	N	5	(B) 12	25	T	6	15
13	N	5	(C) $\frac{1}{2}$				

Strand codes: **N** Number, fractions, percent, ratio · **A** Algebra and patterns · **G** Geometry and measurement · **D** Data and probability · **T** Number theory and counting · **L** Logic. Blueprint: N 6 · A 4 · G 6 · D 3 · T 4 · L 2 items.

Figure notes

Q15 and Q16 are drawn to scale. Q15's missing sides (4 cm and 3 cm) are not labelled on purpose: finding them is part of the item. Q16's angles are 54°, 84° and 42° as drawn.

Partial-credit rules

- **Questions 1-10 (2 points).** Correct letter: 2. Otherwise 0. No partial credit.
- **Questions 11-20 (5 points).** Correct letter: 5. On items with a method box, when the letter is wrong or blank: 3 points for a correct representation with a single arithmetic slip; 2 points for a correct representation that stops or goes wrong later; 0 otherwise. Items without a method box: 5 or 0.
- **Questions 21-25 (6 points).** Correct answer: 6. Otherwise: 2 points for a correct representation, 4 points for a correct representation plus the key intermediate result below; 0 otherwise.
- Score the persistence rubric on its own sheet. Never add or subtract persistence points here.

Q	Correct representation (2)	Adds to 3 (5-pt) / 4 (6-pt)
11	A correct two-step chain (sale price 60, then tax on 60) or $80 \times 0.75 \times 1.13$.	Chain correct but one multiplication slipped.
12	A bar model or ratio table with 8 equal parts totalling 48.	Part size 6 found but final difference slipped.
14	A table of squares vs toothpicks showing +3, or the rule $3n + 1$.	Rule correct, final evaluation slipped.
15	The figure split into two rectangles, or the 10×8 rectangle with the missing 4×3 corner marked.	Correct decomposition, one product or sum slipped.
18	An equation such as $5 \div \text{total} = 1/3$, or $\text{total} = 15$ stated.	Total 15 found, subtraction slipped.
19	A Venn diagram or the three counts 33, 20 and 6.	Counts correct, combination slipped.
21	A bar/ratio model with parts $3k$ and $4k$, or $3k + 6 = 4k$.	$k = 6$ (or 18 boys and 24 girls before).
22	Numbers written as $n - 4$, $n - 2$, n , $n + 2$, $n + 4$ (or n , $n + 2$, ...).	The middle number is 34.
23	A sketch or layer view marking corner, edge and face cubes.	2 two-face cubes on each edge.
24	A 6×6 outcome grid or organised list of ordered pairs.	5 favourable outcomes identified.
25	An organised list or table by hundreds digit.	At least three cases counted correctly (e.g. 1, 2, 3).

Per-item teacher notes

Q	Answer	Concept tested	Likely error	Faster method
1	(D) 27	Fraction of a quantity	(A) 9 — found $1/4$ only	$36 \div 4 = 9$, then $\times 3$.
2	(B) 12	Percent of a quantity	(E) 65 — subtracted 15 from 80	10% is 8, 5% is 4; $8 + 4$.
3	(C) 9	Solving a one-step-then-two-step equation	(D) 12.5 — added 7 instead of subtracting: $50 \div 4$	Undo: $43 - 7 = 36$, $36 \div 4 = 9$.
4	(D) 68	Arithmetic sequence: n th term	(E) 75 — used $5 + 10 \times 7$ (one step too many)	10th term = first + 9 steps = $5 + 63$.
5	(C) 38 cm	Perimeter of a rectangle	(D) 84 cm — found the area	$2 \times (12 + 7)$.
6	(C) 65°	Angle sum of a triangle	(D) 115° — added the two angles and stopped	$180 - 48 - 67 = 180 - 115$.
7	(D) 14	Mean of a data set	(B) 12 — gave the median	Sum 70, $\div 5$.

Q	Answer	Concept tested	Likely error	Faster method
8	(A) 59	Primes; divisibility by 3 and 7	(E) 91 — looks prime, but $91 = 7 \times 13$	Digit sums 12, 6, 12 show 39, 51, 57 are multiples of 3; $91 = 7 \times 13$.
9	(E) 24	Lowest common multiple	(A) 2 — gave the greatest common factor	List multiples of 8 until one is a multiple of 6: 8, 16, 24.
10	(A) Ben	Logical deduction from constraints	(E) — did not combine the two clues	Cat is Cy's; Ben is not dog, so Ben has the fish.
11	(B) \$67.80	Successive percent change	(C) \$70.40 — added -25% and $+13\%$ to get -12% of \$80	$80 \times 0.75 = 60$; $60 \times 1.13 = 67.80$.
12	(B) 12	Ratio with a fixed total (parts of a whole)	(A) 6 — found one part ($48 \div 8$) and stopped	8 parts = 48, so a part is 6; blue – red = 2 parts = 12.
13	(C) $\frac{1}{2}$	Fraction of a remainder ('of what')	(B) $\frac{5}{12}$ — took $\frac{1}{4}$ of the whole, not of the rest	Left after Mia: $\frac{2}{3}$; Leo leaves $\frac{3}{4}$ of that: $\frac{2}{3} \times \frac{3}{4} = \frac{1}{2}$.
14	(B) 61	Linear pattern from a table	(D) 80 — used 4 per square, ignoring shared sides	First square 4, each new square adds 3: $4 + 19 \times 3$.
15	(C) 68	Composite area; finding a missing side	(E) 80 — used the full 10×8 rectangle	$10 \times 8 - (10 - 6) \times (8 - 5) = 80 - 12$.
16	(B) 42	Angles on a straight line; one-variable equation	(C) 63 — solved $2x + 54 = 180$	$3x = 180 - 54 = 126$.
17	(C) 5 cm	Volume of a prism (reverse)	(E) 45 cm — divided by 8 only	Base area 72; $360 \div 72$.
18	(D) 6	Probability as a fraction of the total; reverse	(E) 15 — gave the total, not the green count	Blue is $\frac{1}{3}$ of the total, so total = 15; green = $15 - 9$.
19	(C) 47	Inclusion-exclusion counting	(D) 53 — counted multiples of 15 twice	$33 + 20 - 6$.
20	(D) Dee	Ordering from constraints	(C) Cal — placed Cal before Dee	Chain: Ari, Bo, Dee, then Cal right after Dee.
21	48	Ratio change when one part changes	42 — the total before the 6 boys joined	The gap of 1 part is 6, so 4 parts = 24 girls; now $24 + 24$.
22	38	Consecutive numbers: sum = count $\times$ middle	34 — gave the middle number	Middle = $170 \div 5 = 34$; largest = $34 + 4$.
23	24	Spatial counting on a cube's edges	8 — counted the corner cubes (three faces)	Each of the 12 edges has $4 - 2 = 2$ middle cubes: 12×2 .
24	$\frac{5}{36}$	Outcome space for two dice	$\frac{5}{11}$ — counted sums as equally likely	Pairs (2,6), (3,5), (4,4), (5,3), (6,2): 5 of 36.
25	15	Organised casework; completeness	10 or 21 — missed or double-counted cases (e.g. allowed a leading 0)	Hundreds digit 5, 4, 3, 2, 1 gives $1 + 2 + 3 + 4 + 5$ cases.